Screenshots:

libraryCatalogSystem.java

```java
package libraryCatalog;

import java.util.List;
import java.util.ArrayList;
import java.util.Scanner;

public class libraryCatalogSystem {

    public static void main(String[] args) {
        genericCatalog<libraryItem> libraryCatalog = new genericCatalog<>(); //Creates libraryCatalog object to store all of the information

        Scanner scanner = new Scanner(System.in);
        int choice;
        do { //User interface menu to interact with the catalog system
            System.out.println("Welcome to the library catalog. Please select an option.");
            System.out.println("1. Add a new item to the library");
            System.out.println("2. Remove an item from the library");
            System.out.println("3. View current library catalog");
            System.out.println("4. Exit");

            choice = scanner.nextInt();
            scanner.nextLine();

            switch (choice) {
                case 1:
                    System.out.print("Enter the title: ");
                    String title = scanner.nextLine();

                    System.out.print("Enter the author: ");
                    String author = scanner.nextLine();

                    System.out.print("Enter the item's ID: ");
                    int itemID = scanner.nextInt();
                    scanner.nextLine();

                    libraryItem item = new libraryItem(title, author, itemID); //Takes user-entered information

                    libraryCatalog.addItem(item); //Adds user-entered information to the libraryCatalog
                    break;
                case 2:
                    for (int i = 0; i < libraryCatalog.items.size(); i++) { //Prints out items for the user to see what they want to remove
                        System.out.println(libraryCatalog.items.get(i).getItemID() + ": " + libraryCatalog.items.get(i).getTitle());
                    }

                    System.out.println("Enter the corresponding number for the item you would like to remove from the catalog.");
                    int removedItem = scanner.nextInt(); //User enters the item ID
                    scanner.nextLine();

                    libraryCatalog.removeItem(removedItem); //The removeItem method then takes the user-entered ID
                    break;
                case 3:
                    libraryCatalog.retrieveItem(); //The retrieveItem method displays all of the items stored in libraryCatalog
                    break;
                case 4:
                    System.out.println("Exiting."); //Exits program
                    break;
            }
        } while (choice != 4);
        scanner.close();
    }
}


class genericCatalog <T extends libraryItem> { //Type bound to libraryItem

    public List<T> items;

    public genericCatalog() {
        this.items = new ArrayList<>();
    }

    Scanner scanner = new Scanner(System.in);

    public void addItem(T item) {
        items.add(item);
        System.out.println("Item added successfully.\n");
    }

    public void removeItem(int itemID) {
        for (int i = 0; i < items.size(); i++) {
            if(itemID == items.get(i).getItemID()) {
                items.remove(i);
                System.out.println("Item removed successfully.");
                return;
            }
        }
        System.out.println("Item not found in library catalog."); //Error handling for if the desired item is not in the catalog
    }

    public void retrieveItem() {
        for (int i = 0; i < items.size(); i++) {
            System.out.print(items.get(i).getItemID() + ": " + items.get(i).getTitle() + ", " + items.get(i).getAuthor());
        }
        System.out.println();
    }
}
```

```
97
98
99 class libraryItem {
00
01     private String title;
02     private String author;
03     private int itemID;
04
05     libraryItem(String title, String author, int itemID) { //Constructor
06         this.title = title;
07         this.author = author;
08         this.itemID = itemID;
09     }
10     //Getter methods
11     public String getTitle() {
12         return title;
13     }
14
15     public String getAuthor() {
16         return author;
17     }
18
19     public int getItemID() {
20         return itemID;
21     }
22 }
```

Code:

```
package libraryCatalog;

import java.util.List;

import java.util.ArrayList;

import java.util.Scanner;

public class libraryCatalogSystem {

	public static void main(String[] args) {

		genericCatalog<libraryItem> libraryCatalog = new
genericCatalog<>(); //Creates libraryCatalog object to store all of the
information



		Scanner scanner = new Scanner(System.in);

		int choice;

		do { //User interface menu to interact with the catalog system

			System.out.println("Welcome to the library catalog. Please
select an option.");

			System.out.println("1. Add a new item to the library");

			System.out.println("2. Remove an item from the library");

			System.out.println("3. View current library catalog");
```

```
                System.out.println("4. Exit");


                choice = scanner.nextInt();
                scanner.nextLine();


                switch (choice) {
                      case 1:
                            System.out.print("Enter the title: ");
                            String title = scanner.nextLine();


                            System.out.print("Enter the author: ");
                            String author = scanner.nextLine();


                            System.out.print("Enter the item's ID: ");
                            int itemID = scanner.nextInt();
                            scanner.nextLine();


                            libraryItem item = new libraryItem(title, author,
itemID); //Takes user-entered information


                            libraryCatalog.addItem(item); //Adds user-entered
information to the libraryCatalog
                            break;
                      case 2:
                            for (int i = 0; i < libraryCatalog.items.size();
i++) { //Prints out items for the user to see what they want to remove


System.out.println(libraryCatalog.items.get(i).getItemID() + ": " +
libraryCatalog.items.get(i).getTitle());
```

```
                                }

                                System.out.println("Enter the corresponding
number for the item you would like to remove from the catalog.");
                                int removedItem = scanner.nextInt(); //User
enters the item ID
                                scanner.nextLine();

                                libraryCatalog.removeItem(removedItem); //The
removeItem method then takes the user-entered ID
                                break;
                        case 3:
                                libraryCatalog.retrieveItem(); //The retrieveItem
method displays all of the items stored in libraryCatalog
                                break;
                        case 4:
                                System.out.println("Exiting."); //Exits program
                                break;
                  }
            } while (choice != 4);
            scanner.close();
      }
}
class genericCatalog <T extends libraryItem> { //Type bound to libraryItem

      public List<T> items;

      public genericCatalog() {
            this.items = new ArrayList<>();
```

```
	}

	Scanner scanner = new Scanner(System.in);

	public void addItem(T item) {
		items.add(item);
		System.out.println("Item added successfully.\n");
	}

	public void removeItem(int itemID) {
		for (int i = 0; i < items.size(); i++) {
			if(itemID == items.get(i).getItemID()) {
				items.remove(i);
				System.out.println("Item removed successfully.");
				return;
			}
		}
		System.out.println("Item not found in library catalog."); //Error handling for if the desired item is not in the catalog
	}

	public void retrieveItem() {
		for (int i = 0; i < items.size(); i++) {
			System.out.print(items.get(i).getItemID() + ": " + items.get(i).getTitle() + ", " + items.get(i).getAuthor());
		}
		System.out.println();
	}
}
```

```
class libraryItem {


	private String title;

	private String author;

	private int itemID;


	libraryItem(String title, String author, int itemID) { //Constructor

		this.title = title;

		this.author = author;

		this.itemID = itemID;

	}

	//Getter methods

	public String getTitle() {

		return title;

	}


	public String getAuthor() {

		return author;

	}


	public int getItemID() {

		return itemID;

	}

}
```

Output:

The example used for the output is a CD item. The CD is added to the catalog. Then the catalog is viewed to prove it is added. After that, an attempt to remove the CD is made. However, the ID is invalid so the error is handled by the program within the removeItem() method. Another attempt is then made and is successful. The program is then exited.

```
Welcome to the library catalog. Please select an option.
1. Add a new item to the library
2. Remove an item from the library
3. View current library catalog
4. Exit
1
Enter the title: Star Wars Original Soundtrack
Enter the author: John Williams
Enter the item's ID: 1234
Item added successfully.
Welcome to the library catalog. Please select an option.
1. Add a new item to the library
2. Remove an item from the library
3. View current library catalog
4. Exit
3
1234: Star Wars Original Soundtrack, John Williams
Welcome to the library catalog. Please select an option.
1. Add a new item to the library
2. Remove an item from the library
3. View current library catalog
4. Exit
2
```

```
1234: Star Wars Original Soundtrack
Enter the corresponding number for the item you would like to remove from the catalog.
123
Item not found in library catalog.
Welcome to the library catalog. Please select an option.
1. Add a new item to the library
2. Remove an item from the library
3. View current library catalog
4. Exit
2
1234: Star Wars Original Soundtrack
Enter the corresponding number for the item you would like to remove from the catalog.
1234
Item removed successfully.
Welcome to the library catalog. Please select an option.
1. Add a new item to the library
2. Remove an item from the library
3. View current library catalog
4. Exit
4
Exiting.
```